

Supplementary Material for GRCMI-WPF Hyperparameter Reporting

Compact final-configuration ledger

Overview

This supplementary material reports the final GRCMI-WPF hyperparameter configuration under a unified field layout. Fixed settings shared by all rows are stated once below, so the main ledger only repeats settings that vary across dataset–turbine entries.

Fixed Settings and Aggregation Check

Item	Value
Input look-back window	$L = 36$ steps
Forecast horizon	$H = 12$ steps
PDM spatial patch	11×11 grid
MAPE rule	MAPE ₁₀₀ , computed in the raw-power domain after masking targets below 100 kW
Dataset aggregation	Arithmetic turbine mean for MSE/RMSE/MAE/SDE/ R^2 ; valid-sample-weighted mean for MAPE ₁₀₀ .

Dataset	Turbines	MSE	RMSE	MAE	SDE	MAPE ₁₀₀	R^2
WF1	33	0.1908	0.4347	0.2581	0.4337	30.9530	0.8303
WF2	33	0.2024	0.4485	0.2675	0.4479	34.4025	0.8200
WF3	4	0.2904	0.5386	0.3368	0.5361	36.4160	0.8176
Eolos	1	0.5822	0.7630	0.5480	0.7627	41.7209	0.6829
Kelmarsh	6	0.2898	0.5374	0.3702	0.5363	39.4551	0.8173

Per-Turbine Final Configurations

Table 1: Final GRCMI-WPF hyperparameters for each dataset–turbine entry.

Dataset	Turbine	Tuned/final configuration
WF1	WT00	lr=1.93e-05; wd=1.00e-06; batch=128; hidden=64; layers=3; heads=16; dropout=0.3037; fusion=0.6051
WF1	WT01	lr=3.38e-05; wd=6.32e-05; batch=128; hidden=64; layers=3; heads=16; dropout=0.1327; fusion=0.5917
WF1	WT02	lr=0.001627; wd=0.002039; batch=128; hidden=192; layers=2; heads=8; dropout=0.2952; fusion=9.97e-04
WF1	WT03	lr=2.32e-05; wd=0.001164; batch=128; hidden=128; layers=3; heads=16; dropout=0.3172; fusion=0.6762
WF1	WT04	lr=4.37e-04; wd=5.57e-04; batch=128; hidden=192; layers=3; heads=8; dropout=0.05022; fusion=0.8961
WF1	WT05	lr=0.001797; wd=0.003141; batch=128; hidden=128; layers=4; heads=4; dropout=0.2723; fusion=0.03535
WF1	WT06	lr=3.43e-04; wd=0.002322; batch=128; hidden=64; layers=2; heads=2; dropout=0.4352; fusion=0.3434
WF1	WT07	lr=0.003292; wd=0.002429; batch=128; hidden=192; layers=3; heads=16; dropout=0.1979; fusion=0.07039
WF1	WT08	lr=0.001848; wd=0.001688; batch=128; hidden=64; layers=1; heads=8; dropout=0.4171; fusion=0.3821
WF1	WT09	lr=4.78e-05; wd=0.002107; batch=128; hidden=64; layers=1; heads=8; dropout=0.2195; fusion=0.9879
WF1	WT10	lr=1.38e-05; wd=8.96e-04; batch=128; hidden=64; layers=2; heads=4; dropout=0.4774; fusion=0.3581
WF1	WT11	lr=9.62e-05; wd=1.34e-04; batch=128; hidden=192; layers=1; heads=4; dropout=0.3811; fusion=0.5101
WF1	WT12	lr=0.00147; wd=0.001054; batch=128; hidden=192; layers=2; heads=2; dropout=0.01971; fusion=0.5115
WF1	WT13	lr=0.00122; wd=0.00158; batch=128; hidden=192; layers=3; heads=8; dropout=0.2863; fusion=0.5607
WF1	WT14	lr=2.08e-04; wd=0.001581; batch=128; hidden=128; layers=2; heads=8; dropout=0.07764; fusion=0.5513

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Table 1: Final GRCMI-WPF hyperparameters for each dataset–turbine entry (continued).

Dataset	Turbine	Tuned/final configuration
WF1	WT15	lr=9.16e-04; wd=9.57e-04; batch=128; hidden=128; layers=1; heads=8; dropout=0.01578; fusion=0.2992
WF1	WT16	lr=3.88e-04; wd=0.001681; batch=128; hidden=192; layers=2; heads=4; dropout=0.1341; fusion=0.9202
WF1	WT17	lr=0.001798; wd=0.001229; batch=128; hidden=256; layers=3; heads=16; dropout=0.2495; fusion=0.7523
WF1	WT18	lr=4.98e-04; wd=0.002054; batch=128; hidden=192; layers=3; heads=4; dropout=0.2323; fusion=0.6183
WF1	WT19	lr=7.54e-04; wd=0.001253; batch=128; hidden=256; layers=2; heads=4; dropout=0.01299; fusion=0.4786
WF1	WT20	lr=6.67e-04; wd=0.001661; batch=128; hidden=128; layers=4; heads=2; dropout=0.1258; fusion=6.28e-05
WF1	WT21	lr=0.00263; wd=9.32e-04; batch=128; hidden=64; layers=1; heads=8; dropout=0.04704; fusion=0.5198
WF1	WT22	lr=1.94e-04; wd=0.00163; batch=128; hidden=256; layers=3; heads=2; dropout=0.406; fusion=0.9966
WF1	WT23	lr=3.35e-04; wd=0.00317; batch=128; hidden=64; layers=3; heads=4; dropout=0.3312; fusion=0.9951
WF1	WT24	lr=5.79e-04; wd=0.002234; batch=64; hidden=192; layers=2; heads=4; dropout=0.2932; fusion=0.7787
WF1	WT25	lr=9.69e-04; wd=0.001546; batch=128; hidden=192; layers=2; heads=4; dropout=0.2143; fusion=0.4922
WF1	WT26	lr=0.001141; wd=0.001231; batch=128; hidden=64; layers=2; heads=2; dropout=0.1343; fusion=0.4273
WF1	WT27	lr=9.87e-04; wd=0.001233; batch=128; hidden=64; layers=4; heads=8; dropout=0.4804; fusion=0.03209
WF1	WT28	lr=4.86e-04; wd=9.78e-04; batch=128; hidden=128; layers=3; heads=2; dropout=0.3023; fusion=0.9976
WF1	WT29	lr=0.001512; wd=0.001526; batch=32; hidden=64; layers=4; heads=16; dropout=0.298; fusion=0.06869
WF1	WT30	lr=0.002272; wd=0.001071; batch=128; hidden=128; layers=1; heads=8; dropout=0.3114; fusion=0.9216
WF1	WT31	lr=5.46e-05; wd=8.14e-06; batch=128; hidden=128; layers=3; heads=16; dropout=0.3354; fusion=0.999
WF1	WT32	lr=0.001365; wd=0.001775; batch=64; hidden=256; layers=2; heads=4; dropout=0.4036; fusion=0.1942
WF2	WT00	lr=5.43e-04; wd=0.002932; batch=128; hidden=256; layers=4; heads=16; dropout=0.04741; fusion=0.9999
WF2	WT01	lr=2.92e-05; wd=6.47e-06; batch=128; hidden=128; layers=3; heads=8; dropout=0.4632; fusion=0.4854
WF2	WT02	lr=5.06e-04; wd=0.001598; batch=128; hidden=192; layers=2; heads=2; dropout=0.2672; fusion=0.004043
WF2	WT03	lr=2.06e-04; wd=0.002018; batch=128; hidden=256; layers=2; heads=2; dropout=0.1887; fusion=0.587
WF2	WT04	lr=5.58e-05; wd=0.002842; batch=128; hidden=192; layers=2; heads=8; dropout=0.4411; fusion=0.4734
WF2	WT05	lr=1.75e-04; wd=0.003068; batch=128; hidden=256; layers=3; heads=2; dropout=0.06176; fusion=0.194
WF2	WT06	lr=0.001329; wd=0.001549; batch=64; hidden=128; layers=3; heads=4; dropout=0.4322; fusion=0.9412
WF2	WT07	lr=0.001611; wd=0.001729; batch=128; hidden=192; layers=3; heads=16; dropout=0.3796; fusion=0.211
WF2	WT08	lr=6.06e-04; wd=0.001366; batch=32; hidden=256; layers=4; heads=2; dropout=0.1953; fusion=0.0529
WF2	WT09	lr=8.79e-04; wd=0.00171; batch=128; hidden=256; layers=1; heads=2; dropout=0.09977; fusion=0.08998
WF2	WT10	lr=0.002005; wd=0.002194; batch=128; hidden=256; layers=4; heads=16; dropout=0.1349; fusion=0.1784
WF2	WT11	lr=0.001948; wd=0.002896; batch=128; hidden=256; layers=1; heads=8; dropout=0.2599; fusion=0.007411
WF2	WT12	lr=2.55e-05; wd=1.11e-04; batch=128; hidden=192; layers=2; heads=4; dropout=0.2483; fusion=0.6031
WF2	WT13	lr=1.56e-04; wd=0.001201; batch=128; hidden=256; layers=2; heads=8; dropout=0.1338; fusion=0.7264
WF2	WT14	lr=3.77e-05; wd=4.23e-06; batch=128; hidden=128; layers=2; heads=2; dropout=0.411; fusion=0.682
WF2	WT15	lr=8.82e-04; wd=0.001211; batch=128; hidden=192; layers=4; heads=4; dropout=0.3113; fusion=0.7303
WF2	WT16	lr=8.89e-04; wd=0.001558; batch=128; hidden=64; layers=1; heads=2; dropout=0.1814; fusion=0.4822
WF2	WT17	lr=0.005256; wd=0.002196; batch=128; hidden=192; layers=1; heads=4; dropout=0.2656; fusion=0.9206
WF2	WT18	lr=7.57e-04; wd=4.75e-04; batch=128; hidden=256; layers=4; heads=16; dropout=0.4421; fusion=0.6184
WF2	WT19	lr=0.001011; wd=0.001671; batch=16; hidden=192; layers=1; heads=16; dropout=0.05066; fusion=0.9906
WF2	WT20	lr=0.001961; wd=9.75e-04; batch=128; hidden=64; layers=2; heads=16; dropout=0.4367; fusion=0.3247
WF2	WT21	lr=3.41e-04; wd=8.59e-04; batch=128; hidden=256; layers=4; heads=8; dropout=0.00205; fusion=0.5949
WF2	WT22	lr=1.74e-05; wd=7.82e-06; batch=128; hidden=128; layers=2; heads=2; dropout=0.4309; fusion=0.6475
WF2	WT23	lr=6.30e-04; wd=0.001039; batch=128; hidden=192; layers=4; heads=2; dropout=0.3307; fusion=0.9816
WF2	WT24	lr=1.84e-05; wd=1.53e-06; batch=128; hidden=256; layers=4; heads=8; dropout=0.3325; fusion=0.1496
WF2	WT25	lr=0.002003; wd=0.002116; batch=64; hidden=192; layers=2; heads=8; dropout=0.1064; fusion=0.8346
WF2	WT26	lr=0.006418; wd=3.34e-04; batch=128; hidden=64; layers=4; heads=2; dropout=0.199; fusion=0.6475
WF2	WT27	lr=0.002117; wd=0.002602; batch=128; hidden=256; layers=1; heads=16; dropout=0.2976; fusion=0.4103
WF2	WT28	lr=9.39e-04; wd=0.001422; batch=128; hidden=64; layers=2; heads=16; dropout=0.05744; fusion=0.856
WF2	WT29	lr=8.25e-04; wd=8.89e-04; batch=128; hidden=256; layers=4; heads=2; dropout=0.3199; fusion=0.299
WF2	WT30	lr=5.52e-04; wd=0.002657; batch=128; hidden=128; layers=3; heads=4; dropout=0.1138; fusion=0.435
WF2	WT31	lr=5.15e-05; wd=4.31e-05; batch=64; hidden=192; layers=3; heads=8; dropout=0.1606; fusion=0.8198
WF2	WT32	lr=0.001083; wd=0.001523; batch=128; hidden=64; layers=3; heads=16; dropout=0.4671; fusion=0.2514
WF3	WT00	lr=7.53e-04; wd=1.80e-04; batch=128; hidden=128; layers=3; heads=8; dropout=0.1004; fusion=0.3963
WF3	WT01	lr=2.69e-05; wd=9.56e-04; batch=32; hidden=256; layers=2; heads=4; dropout=0.09242; fusion=0.6576

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Table 1: Final GRCMI-WPF hyperparameters for each dataset–turbine entry (continued).

Dataset	Turbine	Tuned/final configuration
WF3	WT02	lr=4.57e-05; wd=1.02e-05; batch=128; hidden=128; layers=2; heads=4; dropout=0.1471; fusion=0.1742
WF3	WT03	lr=1.13e-05; wd=4.74e-05; batch=16; hidden=64; layers=1; heads=8; dropout=0.1158; fusion=0.7047
Eolos	WT00	lr=0.001984; wd=2.43e-04; batch=128; hidden=64; layers=3; heads=2; dropout=0.09568; fusion=0.1357
Kelmarsh	WT00	lr=0.001554; wd=0.005573; batch=32; hidden=128; layers=1; heads=8; dropout=0.04933; fusion=0.4766
Kelmarsh	WT01	lr=3.47e-04; wd=0.003913; batch=128; hidden=64; layers=2; heads=2; dropout=0.07237; fusion=0.1299
Kelmarsh	WT02	lr=2.10e-04; wd=0.00541; batch=128; hidden=192; layers=2; heads=4; dropout=0.06278; fusion=0.08944
Kelmarsh	WT03	lr=6.71e-04; wd=0.003118; batch=128; hidden=64; layers=1; heads=2; dropout=0.07117; fusion=0.9227
Kelmarsh	WT04	lr=0.001282; wd=0.00248; batch=64; hidden=64; layers=2; heads=8; dropout=0.04986; fusion=0.2176
Kelmarsh	WT05	lr=2.30e-04; wd=5.84e-05; batch=64; hidden=128; layers=2; heads=8; dropout=0.1148; fusion=0.4501